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Hot water immersion: Maintaining core body temperature above 38.5°C mitigates muscle fatigue

Benoît Sautillet¹  | Nicolas Bourdillon² | Grégoire P. Millet²  |
Frédéric Lemaître³ | Maryne Cozette¹  | Stéphane Delanaud⁴ | Saïd Ahmaïdi¹ |
Guillaume Costalat¹ 

¹Faculty of Sport Sciences, APERE Laboratory, UR 3300, University of Picardie Jules Verne, Amiens, France

²Institute of Sport Sciences, University of Lausanne, Lausanne, Switzerland

³Faculty of Sport Sciences, CETAPS Laboratory, UR 3832, Normandy University, Rouen, France

⁴PériTox UMR_I 01 laboratory, CURS-UPJV, F-80054, University of Picardie Jules Verne, Amiens, France

Correspondence

Benoît Sautillet, laboratoire "Adaptations Physiologiques à l'Exercice et Réadaptation à l'Effort" (APERE, UR 3300), UFR STAPS, avenue Paul Claudel, 80025, Amiens cedex 1, France.
Email: benoit.sautillet@u-picardie.fr

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Abstract

Purpose: Hot water immersion (HWI) has gained popularity to promote muscle recovery, despite limited data on the optimal heat dose. The purpose of this study was to compare the responses of two exogenous heat strains on core body temperature, hemodynamic adjustments, and key functional markers of muscle recovery following exercise-induced muscle damage (EIMD).

Methods: Twenty-eight physically active males completed an individually tailored EIMD protocol immediately followed by one of the following recovery interventions: HWI (40°C, HWI₄₀), HWI (41°C, HWI₄₁) or warm water immersion (36°C, CON₃₆). Gastrointestinal temperature (T_{gi}), hemodynamic adjustments (cardiac output [CO], mean arterial pressure [MAP], and systemic vascular resistance [SVR]), pre-frontal cortex deoxyhemoglobin (HHb), ECG-derived respiratory frequency, and subjective perceptual measures were tracked throughout immersion. In addition, functional markers of muscle fatigue (maximal concentric peak torque [T_{peak}]) and muscle damage (late-phase rate of force development [RFD₁₀₀₋₂₀₀]) were measured prior to EIMD (pre-), 24 h (post-24 h), and 48 h (post-48 h) post-EIMD.

Results: By the end of immersion, HWI₄₁ led to significantly higher T_{gi} values than HWI₄₀ (38.8 ± 0.1 vs. 38.0 ± 0.6 °C, $p < 0.001$). While MAP was well maintained throughout immersion, only HWI₄₁ led to increased (HHb) ($+4.2 \pm 1.47$ μ M; $p = 0.005$) and respiratory frequency ($+4.0 \pm 1.21$ breath.min⁻¹; $p = 0.032$). Only HWI₄₁ mitigated the decline in RFD₁₀₀₋₂₀₀ at post-24 h ($-7.1 \pm 31.8\%$; $p = 0.63$) and T_{peak} at post-48 h ($-3.1 \pm 4.3\%$, $p = 1$).

Conclusion: In physically active males, maintaining a core body temperature of ~25 min within the range of 38.5°C–39°C has been found to be effective in improving muscle recovery, while minimizing the risk of excessive physiological heat strain.

KEYWORDS

exercise-induced muscle damage, hemodynamic adjustments, heat dose, muscle recovery, pre-frontal cortex oxygenation, recovery intervention

1 | INTRODUCTION

Exercise-induced muscle damage (EIMD) is a transient phenomenon characterized by structural damage to myofibers and symptoms occurring both immediately and for up to 14 days after the initial bout.¹ As a result, EIMD is often associated with decreased muscle function (i.e., muscle fatigue) and reduced range of joint motion together with the occurrence of delayed onset muscle soreness (DOMS).² In athletes, EIMD monitoring is crucial given its potential to impair performance while increasing the likelihood of injuries.³ A significant concern for many athletes is the ability to recover from EIMD to prime their body for future events. Therefore, athletes who require rapid recovery between bouts of damaging exercise should implement evidence-based recovery strategies to enhance muscle recovery process.

Recovery interventions involving muscle temperature manipulation are believed to improve post-exercise recovery, with cryotherapy being especially popular among athletes and coaches.⁴ Often claimed for its anti-inflammatory properties, the use of cryotherapy to accelerate the recovery of muscle function is nowadays challenged.⁵ In addition, cold water immersion was shown to be as effective—but not better—than a sham condition on muscle recovery, thus questioning the relevance of this strategy.⁶ Conversely, thermotherapy in the form of passive hot water immersion (HWI), hot water-perfused cuff, microwave diathermy, sauna, or heat pad have gained interest as a way to promote muscle recovery, despite current limited data.⁷ In healthy humans, a single bout of thermotherapy with prior exercise-induced fatigue led to immediate improvement in both contractile and neuromuscular function.⁸ Conversely, in the late recovery phase from muscle fatigue or EIMD (i.e., from post-24 h to post-72 h), contrasting outcomes have been observed on functional recovery, muscle damage (creatine kinase and lactate dehydrogenase), or markers relevant to DOMS.^{9–11}

Exercise is a physiological stress factor known to up-regulate heat shock proteins (HSP), including the superfamily HSP70. In the hours and days following EIMD, an accumulation of HSP70 within the muscle has been observed.¹² Recent evidence suggests that these molecular chaperones play a key role in muscle regeneration process in response to muscle damage/injury.¹³ In humans, heat stress per se also appears to promote a heat shock response.¹⁴ Taken together, heat may act as a potent exogenous stress to promote HSP accumulation and thus muscle regeneration.¹⁵ However, an increase in core temperature does not systematically coincide with HSP accumulation as evidenced in the study by Fuchs and colleagues showing that a significant but modest rise in core temperature

($\sim +0.6^{\circ}\text{C}$ from baseline) had no effect on HSP expression.¹⁶ In the above-mentioned studies that failed to show clear benefits of thermotherapy,^{9–11} it is worth noting that the temperature used was within the range of 36°C – 39°C with immersion time from 10 to 30 min, thus preventing any significant rise in core body temperature and subsequent heat shock response.

Although guidelines have recently emerged in the management of sports event in the heat,¹⁷ there is currently no recommendation on the heat dose for recovery purposes. Many of the previous studies relied on passive HWI with the expectation of an improved functional recovery, presumably for its convenience on the field and lower cost compared with other heat-based therapies. For instance, Sabapathy and colleagues found that HWI mitigates the decline in short-term muscle strength and soreness in healthy humans.¹⁸ However, their heat-based protocol—whole-body HWI at 42°C for 45 min—induced a rapid rise in core body temperature up to 39.4°C ($\geq +2^{\circ}\text{C}$), likely explaining the substantial thermal discomfort experienced by the participants. In some cases, these perceptions were further associated with adverse symptoms (e.g., nausea and dizziness) fairly similar to those reported with milder forms of heat-related illness.¹⁹

Taken together, the existing literature suggests a potential dose–response relationship, where low-to-moderate levels of immersion temperature (36°C – 39°C) may have limited effects on muscle recovery, while a more severe immersion temperature (42°C) could yield beneficial outcomes. Based on these findings, it remains to be determined whether an optimal immersion temperature within the range of 39°C and 42°C exists, which can maintain muscular benefits while minimizing the occurrence of clinical symptoms associated with excessive rise in core body temperature. In a previous work, comparison of thermal responses to immersion demonstrated that HWI at 40°C induced a modest rise in core body temperature as opposed to immersion at 41.5°C .²⁰ This suggests that HWI at 40°C may not offer an adequate stimulus to effectively promote accelerated muscle recovery. However, an investigation is warranted to elucidate whether two closely spaced temperatures elicit divergent effects on muscle recovery.

Titration of the optimal heat dose is also crucial to preserve the health of athletes, particularly with regard to maintaining the integrity of the brain. The significant rise in core body temperature ($\geq +2^{\circ}\text{C}$) observed in the study by Sabapathy and colleagues is expected to be safe for the brain parenchyma.²¹ However, recent evidence suggests that such a rise in core body temperature may temporarily disrupt the balance of cerebral metabolic supply and demand, as a consequence of reduced cerebral blood flow.²² The resulting decrease in brain

oxygen delivery may worsen orthostatic intolerance,²³ posing a potential risk when athletes stand up to exit the bath.²⁰

The objective of the present study was to assess the effects of a bearable, practical (lasting <1 h; limiting cardiovascular strain and excessive rise in core body temperature toward thresholds associated with adverse symptoms) and evidence-based heat protocol. Therefore, the main aim of this study was to quantitatively compare, following EIMD, the effects of two close exogenous heat strains (40°C, HWI₄₀, and 41°C, HWI₄₁) on key functional markers of muscle recovery. In addition, gastrointestinal temperature, hemodynamic adjustments, pre-frontal cortex oxygenation, ECG-derived respiratory frequency, and perceptual measures were tracked throughout HWI. We hypothesized that both HWI modalities would increase heat and cardiovascular strain, with HWI₄₁ but not HWI₄₀ mitigating EIMD-induced decline in strength parameters and soreness.

2 | MATERIALS AND METHODS

2.1 | Participants

Thirty healthy males (age: 20.5 ± 2.0 years old; body mass: 69.5 ± 11.2 kg; height: 176.9 ± 6.8 cm) were recruited and split into three groups: HWI₄₁ (41°C, $n = 10$), HWI₄₀ (40°C, $n = 10$), and CON₃₆ (36°C, $n = 10$). Participants were physically active men (>10 h of training per week),

familiar with resistance-type exercise and EIMD with no current illness, no history of lower limb musculoskeletal injuries and no past or present respiratory/cardiac disorders. Participants were asked to refrain from exercising or using any additional recovery strategy throughout the protocol. All individuals completed a medical history, an exercise questionnaire, and a written informed consent to participate. The study received institutional ethical approval and was conducted in accordance with the 1964 Helsinki declaration and its later amendments.

2.2 | Experimental design

Our single-blind, parallel-group, randomized controlled study was designed to primarily compare the effects of two close water temperatures (40°C and 41°C) on key markers of muscle recovery. As depicted in Figure 1, participants visited the laboratory on three consecutive days. Day 1 consisted of assessing baseline measures (pre-) as well as to complete the standardized EIMD protocol. Immediately following EIMD, participants were randomly allocated to one of the following recovery interventions: HWI₄₁, HWI₄₀, or CON₃₆. The CON₃₆ was considered as the sham intervention since this hot water temperature is not expected to significantly increase core body temperature. In the next two visits (post-24 h and post-48 h), the same variables as in Day 1 were measured using the same methodology.

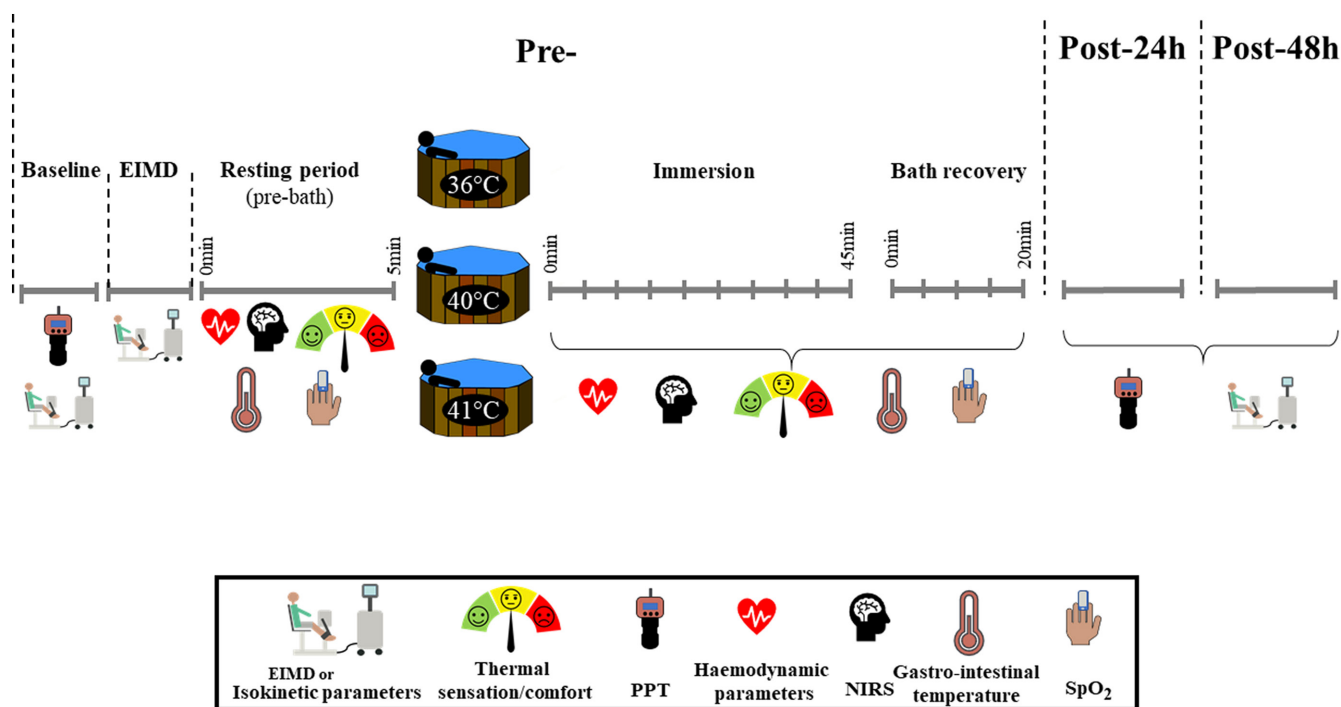


FIGURE 1 Timeline of the experimental design. See text for acronyms.

2.3 | Immersion protocol

Hot water recovery consisted of immersion in an inflatable bath (Novarden, NetSpa Poolstar, France) to the waist level, seated, legs fully extended and forearms non-submerged. Immersion up to the waist level was preferred to whole-body immersion to maximize participants' adherence to the protocol. Water temperature was continuously checked (every 5 min) by means of a water thermometer. If required, hot water coming from the tap (45°C) was used to maintain the target temperature ($\pm 0.1^\circ\text{C}$) and drained meanwhile to ensure its level was kept up to the waist. Participants were invited to drink water (~1 L) throughout immersion and following the bath, a fluid ingestion equivalent to 120% of body mass loss. During HWI₄₀ and HWI₄₁, termination criteria for the immersion included either one of the following: (i) gastrointestinal temperature $\geq 39^\circ\text{C}$ since higher temperature than 39°C may lead to significant heat exhaustion-like symptoms²⁴; (ii) blood pressure thresholds associated with presyncope, that is, a drop in systolic or diastolic of $>20\text{ mmHg}$ or 10 mmHg , respectively; (iii) fatigue, dizziness, palpitations, visual or hearing disturbances, nausea, neck, or back discomfort, and (iv) thermal sensation or thermal comfort rated on the visual scale as "extremely hot" or "extremely uncomfortable," respectively. Unlike relying on termination criteria, the immersion time during CON₃₆ was determined prior to the initiation of the session. This approach was chosen due to the expectation of negligible changes in core body temperature, blood pressure, or subjective markers. Specifically, the immersion time for CON₃₆ was aligned with the mean duration observed during the pre-experimental phase of HWI₄₀ and HWI₄₁ (~38 min). No information about the water temperature nor the research hypothesis was given to the participants to avoid any sham effect.

2.4 | Gastrointestinal temperature

Throughout the recovery interventions, gastrointestinal temperature (T_{gi}) was sampled every 30s using an ingestible telemetric temperature capsule system (e-Celsius, BodyCap). The e-Celsius system has shown to provide good agreement in vivo compared with rectal temperature during exercise-heat stress.²⁵ The capsule was ingested more than 6 h prior to onset of the recovery intervention. Prior to water immersion, participants were invited to drink a small amount of cold water to establish whether the sensor had departed the stomach.

2.5 | Thermal impulse and physiological strain index calculation

Two metrics were used to quantitatively estimate the endogenous heat dose experienced by the participants during the two conditions (HWI₄₁ and HWI₄₀). The thermal impulse represents the area under the curve when core body temperature exceeds 38.5°C , calculated as follows²⁶:

$$T_{i>38.5} = \int (T_{\text{gi}-i} - T_{\text{gi}-38.5}) dt$$

where $T_{i>38.5}$ is the thermal impulse ($^\circ\text{C}\cdot\text{min}$), that is, the area under the curve when T_{gi} exceeds 38.5°C ; $T_{\text{gi}-i}$ is the gastrointestinal temperature at any time i ; $T_{\text{gi}-38.5}$ is the gastrointestinal temperature at 38.5°C ; dt is the duration of each recovery intervention.

In addition, the highest physiological heat strain experienced by the participants during immersion was calculated as follows²⁷:

$$\text{PSI} = 5 \cdot \frac{T_{\text{gi}}(\text{zenith}) - T_{\text{gi}}(0)}{39.5 - T_{\text{gi}}(0)} + 5 \cdot \frac{\text{HR}(\text{zenith}) - \text{HR}(0)}{180 - \text{HR}(0)}$$

where PSI is the physiological strain index (AU); T_{gi} (zenith) and HR (zenith) are the highest gastrointestinal temperature and heart rate experienced by the participants, respectively; $T_{\text{gi}}(0)$ and HR (0) are baseline T_{gi} and HR, respectively.

2.6 | Isokinetic dynamometer and data acquisition

The EIMD protocol and assessment of quadriceps *femoris* function were both performed by means of an isokinetic dynamometer (Cybex Norm®, division of Lumex, Inc.). Before testing, participants followed a standardized warm-up made up of 7 min of cycling at 75 W in addition to three warm-up contractions on the dynamometer. Participants were seated in an upright position and then fastened using straps to secure the chest, hip, and ankle (level arm) of the dominant (working) leg. The lateral epicondyle of the knee joint was aligned with the dynamometer's axis of rotation. Raw dynamometer signals (torque, angular displacement, and velocity) were recorded at 1000 Hz with a PowerLab-16/35 data acquisition system and later analyzed with the LabChart Pro software package (version 8.1, ADInstruments). Prior to data analysis, raw data were filtered (low-pass: 10 Hz) and resampled to 100 Hz.

2.7 | Exercise-induced muscle damage

An individually tailored EIMD protocol was chosen given the wide inter-individual response to a standardized eccentric bout. Therefore, our primary criterion to assess EIMD was based on force-generating capacity rather than a predefined number (sets) of eccentric contractions. A moderate reduction in force-generating capacity was considered in the present study, that is, a 20% to 30% loss in maximal concentric peak torque (T_{peak}) from baseline values. This range has been chosen to ensure significant impairment in muscle strength and pronounced elevation in DOMS-related pain up until the last reassessment of muscle function, that is, post-48 h.²⁸ Briefly, all participants first completed five sets of 10 maximal unilateral, isokinetic ($30^{\circ} \cdot \text{s}^{-1}$), eccentric contractions with their quadriceps *femoris* muscle. The range of motion was set from 0° (full knee extension) to 90° knee flexion for all participants. Then, participants were asked to complete a single set of three maximal, isokinetic, concentric knee extension at $0.02 \pm 0.01 \cdot \text{s}^{-1}$, with the best trial retained to assess reduction in T_{peak} . If strength loss criterion was not met, two to four sets of 10 maximal eccentric contractions were proposed until strength loss reached a value $>20\%$. To limit possible muscle injuries, participants were limited to 90 eccentric contractions. During each contraction, participants were verbally encouraged to resist the movement maximally throughout the full range of motion while having real-time visual feedback of their performance on a computer screen. At the completion of each eccentric contraction, the leg was immediately returned passively to the starting position and another eccentric contraction was immediately initiated.

2.8 | Assessment of muscle function and soreness

In the three conditions, all measurements were assessed at baseline (pre-) and daily throughout the recovery period (post-24 h and post-48 h) as outlined in Figure 1. Two sets of three maximal, concentric knee extension were performed at a slow isokinetic angular velocity ($60^{\circ} \cdot \text{s}^{-1}$) to measure T_{peak} (Nm). The two sets were separated by a 2-min passive recovery. Among the six concentric contractions, the highest T_{peak} value was retained for further analysis. The intra-individual coefficient of variation of this test was $<10\%$.

In addition, the concentric contraction that produced the highest T_{peak} value was used to calculate the rate of force development (RFD), a more representative indirect marker of muscle damage than T_{peak} .²⁹ Given its high reliability and sensitivity to EIMD, the late-phase RFD

($\text{RFD}_{100-200}$, $\text{N} \cdot \text{s}^{-1}$) was preferred to its early phase (e.g., RFD_{0-50}). $\text{RFD}_{100-200}$ was determined as the slope of the force vs. time curve ($\Delta \text{force} / \Delta \text{time}$) within time intervals of 100–200 ms relative to onset of contraction. Onset of muscle contraction was defined as the instant when the knee extensor force exceeded the baseline force by 7.5 N.

Pressure pain threshold (PPT) was assessed by means of a hand-held digital algometer (FDX-50, Wagner Instruments) to quantify the recovery from DOMS. The pressure ($\text{N} \cdot \text{cm}^{-2}$) was applied gradually and perpendicularly to midbelly sites of the *vastus lateralis*. A 1-cm^2 hard rubber tip was used to apply the pressure. The first point at which the first painful sensation was felt by the subject corresponded to PPT. At each time point, three consecutive measurements were done by a single trained examiner (BS) and then averaged for further analysis. For a given subject, a total of nine measurements were done on the same muscle site. Given that each PPT test itself produces exogenous (artificial) pain, a carry-over effect may have occurred throughout the protocol; thus, bilateral measurements (i.e., measures of the control and the tired leg) were performed to ensure that any changes in PPT values were attributable to the recovery intervention.

2.9 | Hemodynamic measurements

Participants were first seated in a neutral room to record a 5-min resting period of physiological and perceptual responses, referred hereafter and in the graphs as the “resting” phase. Cardiac output (CO), stroke volume (SV), and heart rate (HR) were continuously tracked by bio-impedance (PhysioFlow PF-05, Manatec Biomedical) throughout immersion and post-immersion. After shaving and applying a mildly abrasive paste (NuPrep) to the skin, two sets of electrodes, one transmitting and the other receiving, were applied above the supraclavicular fossa (left side) and along the xiphoid process. Another pair of electrodes was used to measure a single electrocardiogram signal (ECG). Electrode contact with the skin was protected from humidity and sweat with adhesive films (Tegaderm™) to preserve the impedance signal.

Simultaneously, systolic (SBP) and diastolic (DBP) blood pressure were measured every 5 min by means of an automatic digital blood pressure monitor (Omron M7 Intelli IT, Omron Healthcare). Mean arterial pressure (MAP) was then obtained from blood pressure recordings, taking into account the increased time dominance of the systolic phase with increasing HR, as follows³⁰:

$$\text{MAP} = \text{DBP} + \left[0.01 \cdot e^{\left(4.14 - \frac{40.74}{\text{HR}} \right)} \right] \cdot \text{PP}$$

where MAP is the mean arterial pressure (mmHg); DBP is the diastolic blood pressure (mmHg); HR is the heart rate (beat.min⁻¹) and PP is the pulse pressure, that is, SBP minus DBP (mmHg).

Finally, systemic vascular resistance (SVR) was obtained as follows, considering the central venous pressure negligible:

$$SVR = \frac{MAP}{CO} \cdot 80$$

where SVR is the systemic vascular resistance (dyne.s.cm⁻⁵); MAP is the mean arterial pressure (mmHg) and CO is the cardiac output (L.min⁻¹).

2.10 | ECG-derived respiratory frequency

A MATLAB-based package provided by Physio Flow's designers was applied to extract ECG recording of each subject. ECG sampling rate was 250 Hz. ECG-derived respiratory frequency (EDR, breath.min⁻¹) is a common and well-accepted technique, for review see.³¹ Briefly, EDR was computed on successive windows of 5 min sliding by 1 min. R and S peaks were identified on the ECG trace and two cubic spline interpolation were performed for the R series and the S series respectively. Amplitude was computed as the difference between the R spline and the S spline. Respiration frequency was estimated from amplitude using a fast Fourier transform (fft, Hamming window, and Welch periodogram) on each window separately.

2.11 | Peripheral oxygenation saturation and pre-frontal cortex oxygenation

Peripheral oxygen saturation (SpO₂) was recorded every 5 min in the bath with a finger probe oximeter (ZuDDo YM101). Simultaneously, pre-frontal cortex oxygenation was assessed continuously using a continuous-wave Near InfraRed Spectrometer (NIRS, OxyMon MK III, Artinis Medical Systems BV). The probe was placed over the forehead, wrapped into a black light-absorbing cloth, and secured with a tensor bandage (around the head) to prevent probe movement and contamination from extraneous light. The NIRS probe setup consisted of one receiver and two transmitters (LED light sources) placed in a row with a source-detector distance of 35 and 45 mm. Each transmitter provides two lights at 857 and 764 nm that are mostly sensitive to changes in oxy- ([HbO₂]) and deoxyhemoglobin ([HHb]) concentrations, respectively. In the present study, only [HHb] is reported

as this NIRS-derived variable appears less sensitive to local blood flow than [HbO₂]. The contribution ratio of the cerebral tissue to the optical signals increases as the source-detector distance widens. Therefore, only [HHb] arising from the source-detector of 45 mm (penetration depth of ~22.5 mm) was considered for further analysis. The differential path-length factor (DPF) was adjusted for each participant. Our NIRS device relies on a continuous-wave technology (i.e., constant DPF), preventing determination of absolute values of [HHb]. Therefore, values are reported as a delta relative to resting values in units of micromolar (μM). The raw data were recorded at a sampling rate of 10 Hz and subsequently downsampled to a rate of 1 Hz for further analysis.

2.12 | Thermal sensation and thermal comfort

Every 5 min into the bath, thermal sensation was assessed using a modified perceptual scale where -5 denotes for "extremely cold," 0 "neutral" and +5 "extremely hot." Similarly, thermal comfort was measured using a modified perceptual scale where 1 denotes for "comfortable" and 5 "extremely uncomfortable."

2.13 | Statistical analysis

A prospective power analysis was used to calculate the required sample size (G*Power software version 3.1). In the present study, a change in T_{peak} across a repeated measure (within-between interaction) was considered as the key outcome variable. The partial eta-squared ($\eta_p = 0.142$) from Sabapathy et al. (2021) was used to calculate the sample size. Based on an assumption of type I error of 0.05, type II error of 0.05 (i.e., power of 95%) and considering a drop-out rate of 20%, 10 participants per group ($n = 30$) were required to determine whether changes were significant.

The samples were first tested for normality, homogeneity of variance, and sphericity with Shapiro-Wilk's, Levene's, and Mauchly's test, respectively. If Mauchly's test was significant, the degrees of freedom were corrected using Greenhouse-Geisser estimates of sphericity. One-way ANOVA with repeated measures was conducted to compare key markers of muscle recovery (T_{peak} , RFD₁₀₀₋₂₀₀, and PPT) across recovery time points (pre- vs. post-24 h vs. post-48 h). Throughout immersion, repeated measures one-way ANOVA was used to track changes in T_{gi} , hemodynamic parameters, [HHb], EDR, SpO₂, and thermal comfort/sensation. Two-way

between-within ANOVA was conducted to compare the above-mentioned variables between the three conditions (between factor) across time points (within factor). Multiple pairwise comparisons adjusted with the Bonferroni correction were conducted whenever the main effect was significant. Statistical analysis and graphing were performed using JASP (version 0.16.10, JASP team, 2022) and OriginPro software (version 2019b, OriginLab), respectively. Statistical significance for all analyses was accepted at $p < 0.05$. Effect sizes associated with ANOVA (omega-squared, ω^2) and subsequent particular comparison (Cohen's d) are reported in the results section. Changes reported in the results section refer to means $\pm$ SD.

3 | RESULTS

3.1 | EIMD-induced muscle fatigue

Two participants were removed from analysis since they did not satisfy strength loss criterion (i.e., a drop in $T_{\text{peak}} < 20\%$ post-EIMD) despite 90 eccentric contractions. Among the remaining participants ($n = 28$), EIMD induced a similar acute muscle fatigue (i.e., drop in T_{peak} values) across the three conditions (CON₃₆: $-21.6 \pm 4.5\%$; HWI₄₁: $-20.3 \pm 7.5\%$; HWI₄₀: $-26.4 \pm 5.6\%$; $p = 0.164$) (Figure 2A).

3.2 | Markers of muscle recovery (T_{peak} , RFD_{100–200}, and PPT)

Changes in T_{peak} and RFD_{100–200} in response to the three conditions are displayed in Figure 2A,B. One-way repeated measures ANOVA revealed a significant effect of condition on T_{peak} across recovery time points (CON₃₆: $F_{(2,14)} = 15.57$; $p < 0.01$; $\omega^2 = 0.40$; HWI₄₁: $F_{(2,12)} = 7.79$; $p = 0.007$; $\omega^2 = 0.04$; HWI₄₀: $F_{(2,12)} = 21.97$; $p < 0.001$; $\omega^2 = 0.19$). More precisely, T_{peak} significantly dropped at post-24 h and post-48 h in response to CON₃₆ and HWI₄₀ (all $p < 0.05$). Conversely, while T_{peak} was significantly depressed at post-24 h in response to HWI₄₁ ($p = 0.008$; $d = 0.48$), no difference was observed at post-48 h when compared to baseline values ($p = 1$; $d = 0.13$). One-way repeated measures ANOVA revealed a significant effect of CON₃₆ ($F_{(2,14)} = 5.04$; $p = 0.022$; $\omega^2 = 0.14$) and HWI₄₀ ($F_{(2,12)} = 4.69$; $p = 0.031$; $\omega^2 = 0.12$) on RFD_{100–200} across recovery time points. More precisely, at any recovery time points, RFD_{100–200} significantly dropped in response to these two conditions (all $p < 0.05$). Conversely, no significant main effect of HWI₄₁ on RFD_{100–200} was observed across recovery time points ($F_{(2,12)} = 0.47$; $p = 0.63$; $\omega^2 = 0$).

Changes in PPT in response to the three conditions are presented in Figure 3. For the control leg, neither CON₃₆, HWI₄₀, nor HWI₄₁ had a significant effect on PPT (all $p > 0.05$). For the tired leg, one-way repeated measures ANOVA revealed a significant effect of CON₃₆ ($F_{(2,14)} = 17.18$; $p < 0.01$; $\omega^2 = 0.15$), HWI₄₁ ($F_{(2,12)} = 6.37$; $p = 0.013$; $\omega^2 = 0.16$), and HWI₄₀ ($F_{(2,12)} = 11.11$; $p = 0.002$; $\omega^2 = 0.41$) on PPT across recovery time points. More precisely, at any recovery time points, PPT significantly dropped in response to CON₃₆ and HWI₄₀ (all $p < 0.05$). Conversely, while PPT was significantly reduced at post-24 h in response to HWI₄₁ ($p = 0.027$; $d = 1.01$), no difference was observed at post-48 h when compared to baseline values ($p = 0.970$; $d = 0.01$).

3.3 | Exogenous and endogenous heat doses

Exogenous and endogenous heat doses experienced by the participants throughout CON₃₆, HWI₄₁, and HWI₄₀ are displayed in Table 1. Termination criteria were achieved slightly earlier during HWI₄₀ compared with HWI₄₁; however, when considering all three conditions together, the duration of immersion was found to be similar ($p > 0.05$). No clinical symptoms occurred either during or after immersion. As expected, there was a significant effect of condition on water temperature ($F_{(2,22)} = 6293.36$, $p < 0.001$, $\omega^2 = 1$) while other key markers describing exogenous heat dose (ambient temperature and relative humidity) were similar across conditions (all $p > 0.05$).

Kinetics of T_{gi} throughout the three conditions are displayed in Figure 4A. Prior to immersion, resting T_{gi} values were similar across the three conditions ($p = 1$). One-way repeated measures ANOVA revealed a significant main effect of CON₃₆ on T_{gi} across time points ($F_{(1,7)} = 7.55$; $p = 0.007$; $\omega^2 = 0.05$); overall, mean T_{gi} slightly decreased in response to CON₃₆ compared with respective resting values ($-0.19^\circ\text{C} \pm 0.25^\circ\text{C}$; $p = 0.028$). There was also a significant effect of HWI₄₁ ($F_{(1,7)} = 145.71$; $p = 0.001$; $\omega^2 = 0.80$) and HWI₄₀ ($F_{(1,7)} = 6.27$; $p = 0.026$; $\omega^2 = 0.22$) on T_{gi} across time points, with both conditions showing a gradual rise in T_{gi} throughout immersion. However, the significant interaction from two-way ANOVA ($F_{(2,21)} = 29.09$; $p < 0.001$; $\omega^2 = 0.37$) and subsequent particular comparisons revealed that the increase in T_{gi} was more pronounced during HWI₄₁ than HWI₄₀ by the end of immersion (38.76 ± 0.13 vs. $37.96^\circ\text{C} \pm 0.62^\circ\text{C}$, $p < 0.001$, $d = 2.88$) until the first recovery time point (38.74 ± 0.24 vs. $37.78^\circ\text{C} \pm 0.61^\circ\text{C}$; $p < 0.001$, $d = 2.67$). Finally, there was a significant effect of thermotherapy on $T_{i>38.5}$ ($F_{(2,19)} = 8.88$; $p = 0.002$; $\omega^2 = 0.41$), with HWI₄₁ showing a significantly higher $T_{i>38.5}$ than HWI₄₀ ($p = 0.009$).

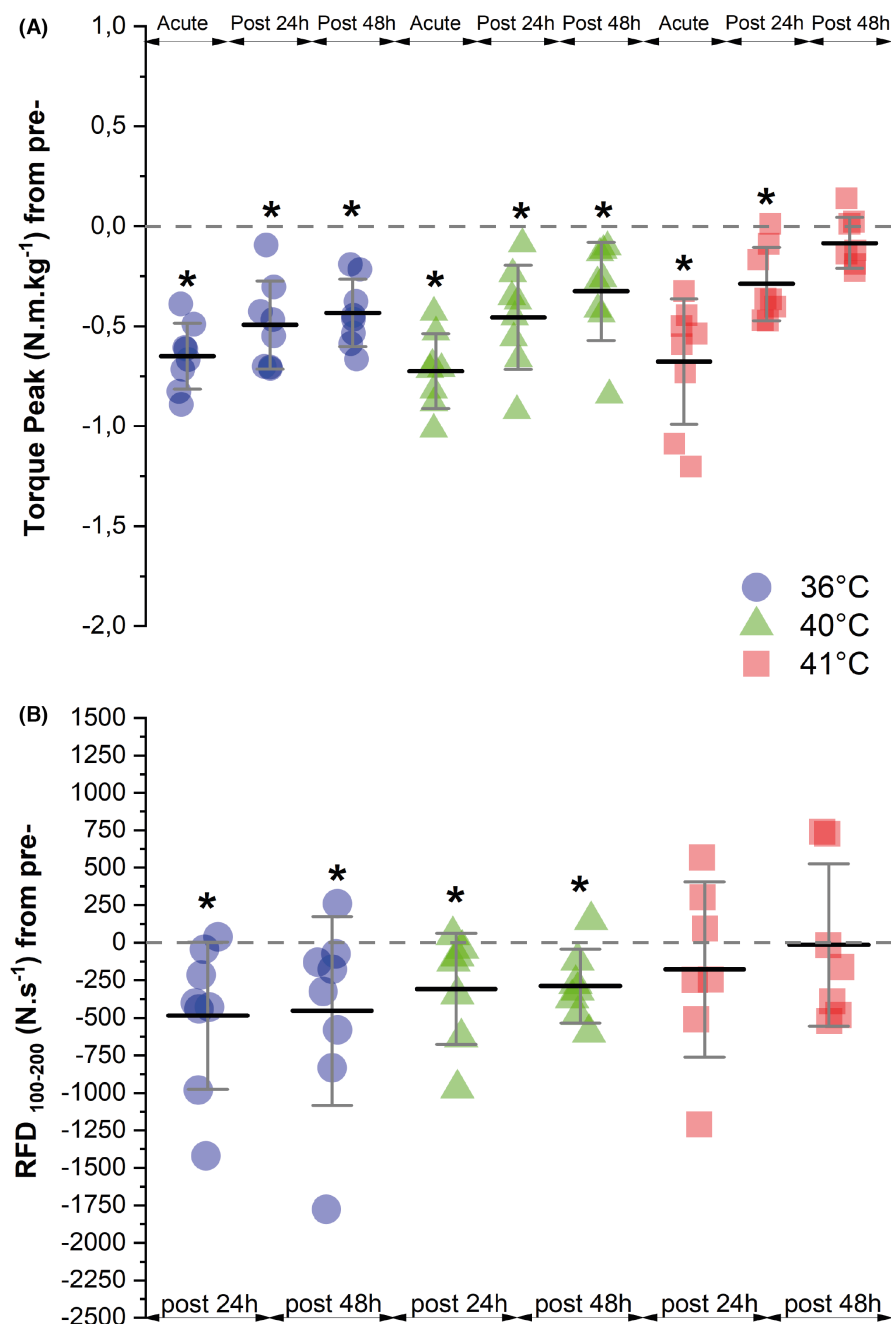


FIGURE 2 Markers of muscle recovery: Individual data points and mean changes from pre- ($\pm$ SD) for torque peak (A), late-phase rate of force development (RDF₁₀₀₋₂₀₀, B) in response to HWI₄₁ (41°C), HWI₄₀ (40°C), or warm water immersion (36°C). In panel A, note that the “acute” phase refers to the acute fatigue (i.e., drop in T_{peak}) immediately following EIMD. * $p < 0.05$ indicates a significant difference from pre-.

3.4 | Hemodynamic parameters

Kinetics of HR, MAP, CO, SVR, and SV throughout the three conditions are displayed in Figure 4B–F.

No significant main effect of CON₃₆ was observed on HR across time points ($F_{(6,36)} = 2.05$; $p = 0.085$; $\omega^2 = 0.01$). However, there was a significant effect of HWI₄₁ ($F_{(5,35)} = 74.75$; $p < 0.001$; $\omega^2 = 0.76$) and HWI₄₀ ($F_{(5,30)} = 26.45$; $p < 0.001$; $\omega^2 = 0.39$) on HR across time points, with both conditions showing a gradual rise in HR throughout immersion. More precisely, HR at the end immersion was significantly increased from resting values in both HWI₄₁ ($+52.5 \pm 12.56$ bpm; $p < 0.001$, $d = 5.27$) and HWI₄₀ ($+32.6 \pm 15.0$ bpm; $p < 0.001$, $d = 2.41$).

However, the significant interaction from two-way ANOVA ($F_{(5,65)} = 5.97$; $p < 0.001$; $\omega^2 = 0.08$) and subsequent particular comparisons revealed that increase in HR was more pronounced during HWI₄₁ than HWI₄₀ by the end of immersion (127.8 ± 13.43 vs. $102.1^\circ\text{C} \pm 17.50^\circ\text{C}$, $p = 0.012$, $d = 2.19$).

Neither CON₃₆ ($F_{(6,36)} = 2.22$; $p = 0.063$; $\omega^2 = 0.04$) nor HWI₄₀ ($F_{(6,36)} = 0.42$; $p = 0.864$; $\omega^2 = 0$) had a significant effect on MAP across immersion time points. However, a significant effect of HWI₄₁ was observed across immersion time points ($F_{(6,42)} = 3.92$; $p = 0.003$; $\omega^2 = 0.05$) with subsequent particular comparisons revealing that MAP only decreased at the time point representing 40% of immersion duration (-7.6 ± 6.44 mmHg, $p = 0.021$, $d = 0.76$).

FIGURE 3 Markers of soreness recovery: Individual data points and mean changes from pre- ($\pm$ SD) for pain-pressure threshold of the control and tired leg in response to HWI₄₁ (41°C), HWI₄₀ (40°C), or warm water immersion (36°C). * $p < 0.05$ indicates a significant difference from pre-.

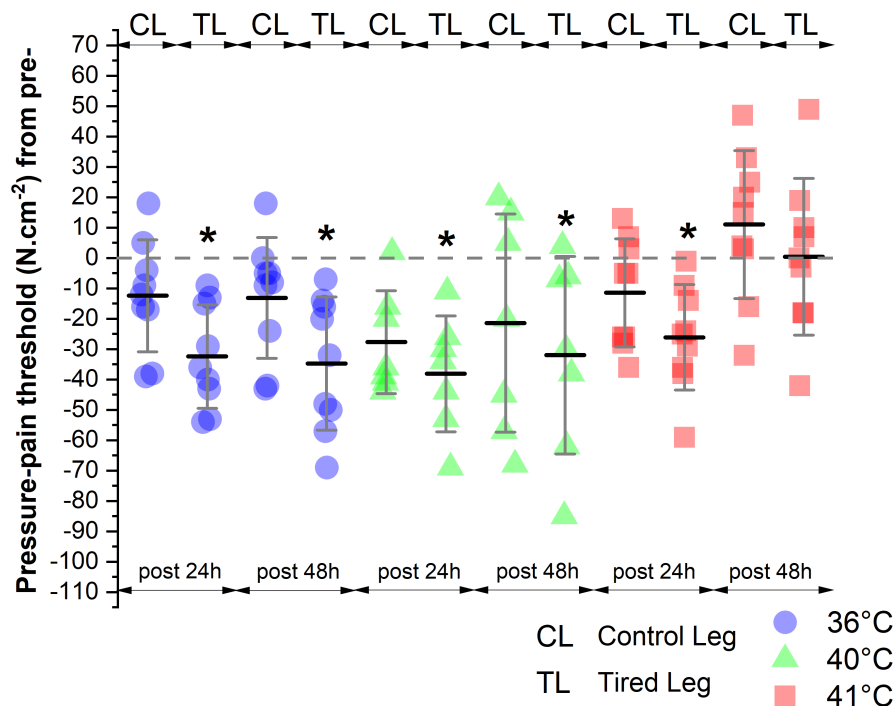


TABLE 1 Exogenous and endogenous heat doses experienced by the athletes.

	Athletes ($n = 28$)		
	CON ₃₆ ($n = 10$)	HWI ₄₀ ($n = 9$)	HWI ₄₁ ($n = 9$)
Exogenous heat dose			
Water T° (°C)	36.2 $\pm$ 0.11	40.0 $\pm$ 0.10*	41.1 $\pm$ 0.08* [#]
Bath duration (min)	38.3 $\pm$ 6.05	42.0 $\pm$ 10.74	47.4 $\pm$ 7.40
Ambient T° (°C)	22.4 $\pm$ 0.36	22.3 $\pm$ 0.20	22.6 $\pm$ 0.43
Ambient relative humidity (%)	87.9 $\pm$ 1.79	88.9 $\pm$ 2.41	89.3 $\pm$ 2.06
Endogenous heat dose			
Zenith T_{gi} (°C)	–	38.02 $\pm$ 0.50*	38.76 $\pm$ 0.31* [#]
ΔT_{gi} (°C)	–0.29 $\pm$ 0.24	0.58 $\pm$ 0.48*	1.32 $\pm$ 0.17* [#]
Time spent > 38.5°C (min)	0 $\pm$ 0	6.0 $\pm$ 11.48	25.5 $\pm$ 8.36 [#]
Thermal impulse (°C.min) (>38.5°C)	0 $\pm$ 0	2.56 $\pm$ 6.68	14.9 $\pm$ 8.72* [#]
Zenith physiological strain index (AU)	0.15 $\pm$ 0.42	3.81 $\pm$ 1.19*	5.93 $\pm$ 0.96* [#]

Note: Values are mean $\pm$ SD.

Abbreviations: CON₃₆, warm water immersion (36°C); HWI₄₀, hot water immersion (40°C); HWI₄₁, hot water immersion (41°C); T° , temperature; T_{gi} , gastrointestinal temperature.

* $p < 0.05$ indicates that HWI₄₀ and HWI₄₁ were significantly different from CON₃₆.

[#] $p < 0.05$ indicates that HWI₄₁ was significantly different from HWI₄₀.

No significant main effect of CON₃₆ was observed on CO across time points ($F_{(5,30)} = 0.38$; $p = 0.856$; $\omega^2 = 0$). However, there was a significant effect of HWI₄₁ ($F_{(1,10)} = 17.23$; $p < 0.001$; $\omega^2 = 0.50$) and HWI₄₀ ($F_{(5,30)} = 20.69$; $p < 0.001$; $\omega^2 = 0.48$) on CO across time points, with both conditions showing a gradual rise in CO throughout immersion. However, no interaction was found for CO between HWI₄₁ and HWI₄₀

across immersion time points ($F_{(5,65)} = 1.73$; $p = 0.141$; $\omega^2 = 0.02$).

One-way repeated measures ANOVA revealed a significant main effect of CON₃₆ on SVR across immersion time points ($F_{(6,36)} = 11.34$; $p < 0.001$; $\omega^2 = 0.17$); overall, mean SVR increased post-CON₃₆ compared with resting values ($+259 \pm 209$ dyn.s.cm⁻⁵; $p = 0.017$; $d = 1.24$) while no difference was observed during immersion (all $p > 0.05$).

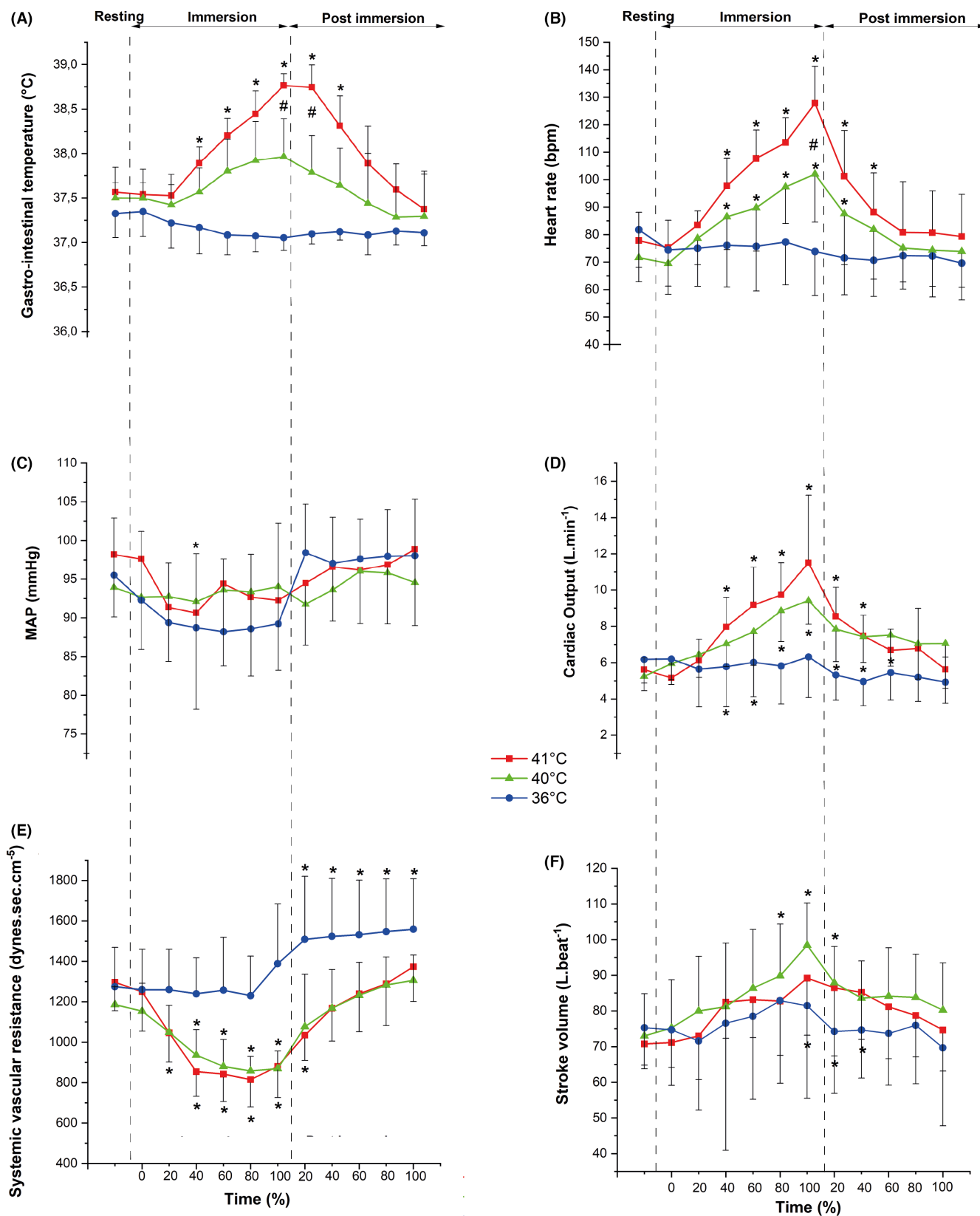


FIGURE 4 Markers of endogenous heat strain and cardiovascular strain: Mean ($\pm$ SD) gastrointestinal temperature (A), heart rate (B), mean arterial pressure (MAP, C), cardiac output (D), systemic vascular resistance (E), and stroke volume (F) in response to HWI₄₁ (41°C), HWI₄₀ (40°C), or warm water immersion (36°C). * $p < 0.05$ indicates a significant difference from resting values. # $p < 0.05$ indicates a significant difference between HWI₄₁ and CON₃₆.

There was also a significant effect of HWI_{41} ($F_{(6,42)} = 35.69$; $p < 0.001$; $\omega^2 = 0.65$) and HWI_{40} ($F_{(6,36)} = 17.02$; $p < 0.001$; $\omega^2 = 0.55$) on SVR across time points, with both conditions showing a gradual decrease in SVR throughout immersion. However, no interaction was found for SVR between HWI_{41} and HWI_{40} across immersion time points ($F_{(5,65)} = 1.98$; $p = 0.093$; $\omega^2 = 0.03$).

Neither CON_{36} ($F_{(5,30)} = 0.76$; $p = 0.59$; $\omega^2 = 0$) nor HWI_{41} ($F_{(5,35)} = 3.52$; $p = 0.011$; $\omega^2 = 0.16$) had a significant effect on SV across immersion time points. However, a significant effect of HWI_{40} was observed across immersion time points ($F_{(5,30)} = 8.69$; $p = 0.001$; $\omega^2 = 0.18$) with subsequent particular comparisons revealing that SV only increased by the end of immersion ($+23.2 \pm 13.4 \text{ mL} \cdot \text{beat}^{-1}$, $p < 0.001$, $d = 1.54$).

3.5 | SpO₂, pre-frontal cortex [HHb], and ECG-derived respiratory frequency

Pre-frontal cortex [HHb], SpO_2 , and EDR during the three conditions are displayed in Figure 5A,B.

Neither CON_{36} ($F_{(5,35)} = 0.30$; $p = 0.873$; $\omega^2 = 0$), HWI_{41} ($F_{(5,40)} = 1.12$; $p = 0.368$; $\omega^2 = 0.01$), nor HWI_{40} ($F_{(5,30)} = 2.47$; $p = 0.054$; $\omega^2 = 0.10$) had a significant effect on SpO_2 across immersion time points.

Neither CON_{36} ($F_{(5,41)} = 0.47$; $p = 0.794$; $\omega^2 = 0$) nor HWI_{40} ($F_{(5,29)} = 0.57$; $p = 0.723$; $\omega^2 = 0$) had a significant effect on [HHb] across immersion time points. However, a significant effect of HWI_{41} was observed across immersion time points ($F_{(5,35)} = 3.46$; $p = 0.012$; $\omega^2 = 0.23$)

with subsequent particular comparisons revealing that [HHb] increased at the end of immersion ($+4.2 \pm 1.47 \mu\text{M}$; $p = 0.005$, $d = 2.18$). In addition, the significant interaction from two-way ANOVA ($F_{(5,10)} = 2.92$; $p = 0.003$; $\omega^2 = 0.06$) and subsequent particular comparisons revealed that [HHb] was higher during HWI_{41} compared with CON_{36} ($p = 0.026$, $d = 2.1$).

Neither CON_{36} ($F_{(1,7)} = 2.64$; $p = 0.148$; $\omega^2 = 0.02$) nor HWI_{40} ($F_{(1,7)} = 0.06$; $p = 0.815$; $\omega^2 = 0$) had a significant main effect on EDR. Conversely, HWI_{41} had a significant main effect on EDR across time points ($F_{(1,7)} = 7.10$; $p = 0.032$; $\omega^2 = 0.24$). Overall, mean EDR throughout HWI_{41} significantly increased from resting values (19.21 ± 3.85 vs. $15.16 \pm 2.63 \text{ breath} \cdot \text{min}^{-1}$, respectively; $p = 0.032$; $d = 1.22$).

3.6 | Thermal comfort and thermal sensation

Kinetics of thermal comfort and thermal sensation throughout the three conditions are displayed in Figure 6A,B.

No significant main effect of CON_{36} was observed on thermal comfort throughout immersion ($F_{(5,30)} = 0.77$; $p = 0.576$; $\omega^2 = 0$). Conversely, there was a significant main effect of HWI_{41} ($F_{(5,40)} = 44.58$; $p = 0.001$; $\omega^2 = 0.68$) and HWI_{40} ($F_{(5,30)} = 19.49$; $p = 0.001$; $\omega^2 = 0.50$) on thermal comfort across time points, with both conditions showing a gradual rise throughout immersion. More precisely, thermal comfort increased from time points representing

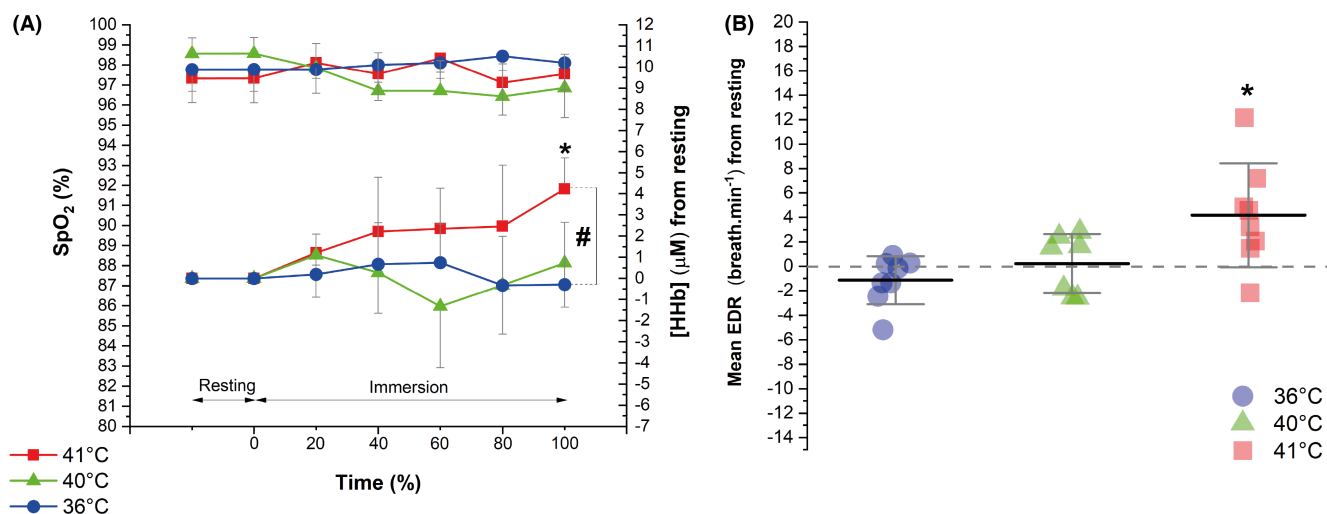


FIGURE 5 Mean ($\pm$ SD) of pre-frontal deoxyhemoglobin ([HHb], A) and peripheral oxygen saturation (SpO_2 , A). Individual data points and mean changes from resting ($\pm$ SD) for ECG-derived respiratory frequency (EDR, B) during HWI_{41} (41°C), HWI_{40} (40°C), or warm water immersion (CON_{36}). Note that unlike other measured hemodynamic parameters, post-immersion [HHb] kinetics are not displayed since the NIRS probe had to be removed due to significant discomfort in most participants. Panel A: * $p < 0.05$ indicates a significant difference from resting. # $p < 0.05$ indicates a significant difference between HWI_{41} and CON_{36} . Panel B: * $p < 0.05$ indicates a significant difference from resting values.

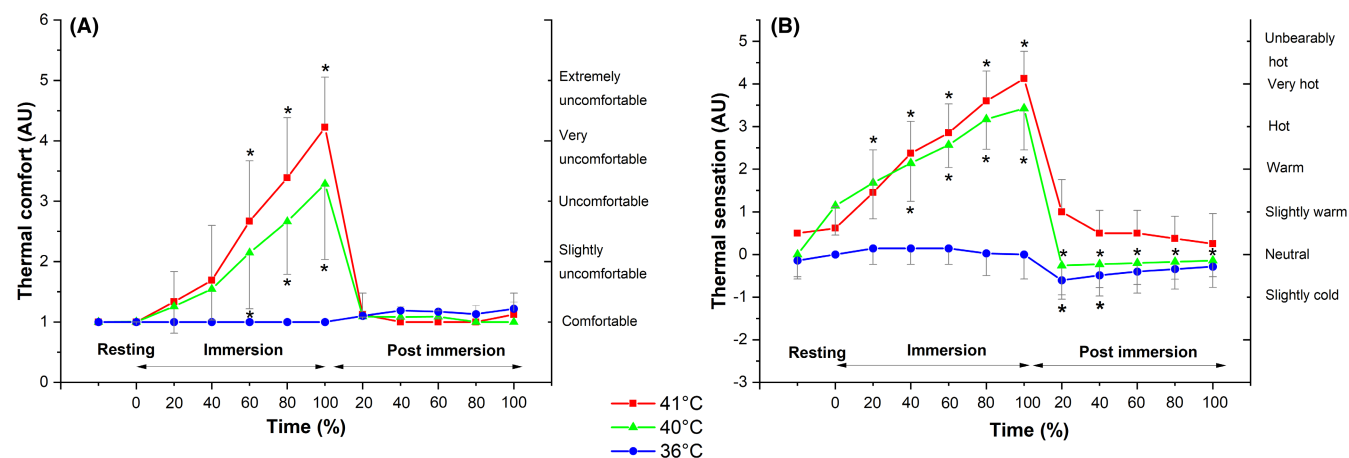


FIGURE 6 Markers of subjective perception: Mean ($\pm$ SD) thermal comfort (A) and thermal sensation (B) in response to HWI₄₁ (41°C), HWI₄₀ (40°C), or warm water immersion (CON₃₆). * $p < 0.05$ indicates a significant difference from resting values.

60% of immersion duration until the end of immersion in both HWI₄₁ and HWI₄₀ (all $p < 0.05$). However, no interaction was found for thermal comfort between HWI₄₁ and HWI₄₀ across immersion time points ($F_{(5,70)} = 1.92$; $p = 0.102$; $\omega^2 = 0.02$).

No significant main effect of CON₃₆ was observed on thermal sensation throughout immersion ($F_{(5,30)} = 0.66$; $p = 0.660$; $\omega^2 = 0$). Conversely, there was a significant effect of HWI₄₁ ($F_{(5,30)} = 39.59$; $p = 0.001$; $\omega^2 = 0.72$) and HWI₄₀ ($F_{(5,30)} = 22.14$; $p = 0.001$; $\omega^2 = 0.50$) on thermal sensation across time points, with both conditions showing a gradual rise throughout immersion. More precisely, thermal sensation increased from time points representing 20% of immersion duration until the end of immersion in both HWI₄₁ and HWI₄₀ (all $p < 0.05$). An interaction was found for thermal sensation between HWI₄₁ and HWI₄₀ across immersion time points ($F_{(5,65)} = 2.50$; $p = 0.039$; $\omega^2 = 0.05$); however, subsequent particular comparisons did not reveal any statistical differences (all $p > 0.05$).

4 | DISCUSSION

A single HWI₄₁ session, rather than CON₃₆ or HWI₄₀, provided the necessary level of physiological challenge to mitigate the decline in muscle strength and soreness following exercised-induced muscle damage. By the end of immersion, HWI₄₁ led to higher gastrointestinal temperature values than HWI₄₀. While mean arterial pressure was well maintained throughout immersion, only HWI₄₁ led to an increase in pre-frontal cortex [HHb] with a concomitant rise in breathing frequency.

Our individually tailored EIMD protocol successfully induced quadriceps *femoris* fatigue, since T_{peak} was significantly reduced in response to sham (i.e., CON₃₆) at post-24 h and post-48 h (Figure 2A). In parallel, both RFD₁₀₀₋₂₀₀

and PPT of the tired leg were decreased within the same period in response to CON₃₆ (Figures 2B and 3). As both these markers (i.e., RFD₁₀₀₋₂₀₀ and PPT) are considered indirect markers of muscle damage,^{29,32} it is reasonable to conclude that our protocol effectively induced muscle damage in all subjects.

To date, focus has been placed on the early recovery phase (i.e., within hours post-exercise) or the chronic responses to heat (i.e., repeated exposures over several weeks). Taken together, these studies demonstrated positive skeletal muscle adaptations such as increased glycogen resynthesis, improved neuromuscular coupling, and hypertrophy while dampening atrophy-related signaling factors.^{7,8,13} The present results highlighted that heat is also relevant in the late recovery stage (24-48 h) to mitigate the decline in functional variables and muscle soreness following EIMD. However, it is worth noting that the responses to two close exogenous heat doses had divergent effects: the key markers of muscle recovery (i.e., T_{peak} and RFD₁₀₀₋₂₀₀) returned to baseline values only in HWI₄₁ and not in both HWI₄₀ and CON₃₆ (Figure 2A,B). The mechanisms underpinning these findings likely involved the expression of HSP given its crucial role in muscle repair following EIMD.¹³ Evidence suggests that HSP upregulation is tightly linked to the magnitude of increase in core body temperature.³³ The most striking endogenous difference between the two hot baths was their impact on T_{gi} kinetic, a well-established proxy for core body temperature. A progressive rise in T_{gi} was observed throughout HWI₄₀ and HWI₄₁; however, it is clear that participants from the HWI₄₁ condition achieved a significantly higher T_{gi} (~38.8°C) than in HWI₄₀ (~38.0°C) (Figure 4A; Table 1). In humans, it was suggested that a target core temperature of 38.5°C is required to magnify HSP70 translation/transcription.³⁴ This dose-response relationship was confirmed in later studies.^{33,35} Using an exercise-heat model, Gibson

and colleagues demonstrated that achieving a core body temperature near—albeit lower than 38.5°C (38.35°C)—had no effect on HSP plasma concentration; conversely, only the condition overcoming the aforementioned (temperature) threshold was associated with significant HSP blood accumulation.³⁵ Our results revealed that HWI₄₁ but not HWI₄₀ induced the required thermal strain to achieve a sustained (~25 min) core temperature > 38.5°C. In line with this finding, the thermal impulse—a more representative index of total heat strain and strong predictor of HSP mRNA changes³³—only increased during HWI₄₁ (Table 1). Although HSP was not measured in the present study, one may speculate that only HWI₄₁ induced HSP accumulation, ultimately promoting the faster muscle recovery following HWI₄₁. Further studies are essential to clarify whether this temperature threshold applies to local HSP responses, as systemic HSP levels may not accurately reflect the HSP activity within skeletal muscle.

The beneficial effects reported on muscle function are in agreement with a recent heat-preconditioning study in healthy active males.¹⁸ However, one of the main difference with our study is the endogenous heat strain (i.e., core body temperature) achieved by our participants which was lower than in the study of Sabapathy and colleagues (38.8°C vs. 39.4°C, respectively). In our view and as stated by others, the magnitude of hyperthermia that is ethically permissible should not exceed a core body temperature of 39°C.²⁴ A hot (and humid) environment will pose a significant thermoregulatory, cardiovascular, and perceptual challenge for an athlete, thereby increasing the likelihood of experiencing heat illnesses, especially in individuals unaccustomed to such physiological challenge.¹⁷ Subjective markers of heat strain (thermal comfort/sensation) were similar between HWI₄₀ and HWI₄₁ across all time points examined (Figure 6A,B), despite significant differences in T_{gi} between the two conditions. Consequently, markers based on subjective perception were not adequately sensitive to differentiate between HWI₄₁ and HWI₄₀. By contrast, HWI₄₁ led to higher HR and T_{gi} values than HWI₄₀, thus explaining the higher physiological strain index (PSI) during HWI₄₁ (5.9) compared with HWI₄₀ (3.8) by the end of immersion (Table 1). This metric—a more objective marker of endogenous heat stress than thermal sensation/comfort—demonstrated that the participants from HWI₄₁ experienced a higher cardiovascular/heat strain than the participants from HWI₄₀. PSI is used to classify an individual's response to a given heat challenge, with 0 representing no physiological strain, 10 representing the highest physiological strain and values ≥ 7.5 being considered as at risk of reaching thermal tolerance limit.³⁶ Participants involved in the study by Sabapathy and colleagues likely achieved a PSI above this “critical” cutoff threshold (i.e., 7.5) since heat exhaustion-like symptoms such as nausea

and dizziness were reported. Conversely, in the present study, none of the enrolled participants complained from significant adverse symptoms while in HWI₄₁, possibly due to the reduced PSI (<7.5). Taken together, our results indicated that a sustained core body temperature of ~25 min within 38.5°C and 39°C is relevant to improve key markers of muscle recovery in healthy physically active males while limiting the likelihood of experiencing excessive physiological heat strain.

Heat exposure represents one of the most substantial challenges to integrative cardiovascular function. CO increased in both HWI₄₁ and HWI₄₀ by the end of immersion (up to twofold, Figure 4D), in line with previous data of passive heating in young, healthy participants.³⁷ It is now well established that the increased CO is primarily driven by an increased HR, as SV is maintained or slightly increased, despite profound reduction in cardiac preload during thermal stress.³⁷ Consistently, in the present study, the rise in HR explained most of the CO response, as SV was either maintained (HWI₄₁) or increased (+36%, HWI₄₀) throughout immersion (Figure 4F). In both conditions, the increase in CO offset the large decrease in SVR (Figure 4E) to maintain MAP (Figure 4C).³⁷ In parallel, NIRS revealed that pre-frontal cortex [HHb] was significantly increased by the end of HWI₄₁ while no such outcome was observed during HWI₄₀ (Figure 5A). Mechanistically, this microvascular O₂ delivery-to-utilization mismatch most likely evokes a compromised blood perfusion in the area of NIRS interrogation rather than a diminished arterial O₂ content, since SpO₂ remained constant throughout HWI₄₁ (Figure 5A). This phenomenon is unlikely due to systemic hypotension since MAP was well maintained throughout HWI₄₁ (Figure 4C). Alternatively, we found that breathing frequency increased during HWI₄₁ but not during HWI₄₀ nor CON₃₆, thus evoking a hyperventilatory response (i.e., increased minute ventilation) while in HWI₄₁ (Figure 5B). Hyperventilation-induced (arterial) hypocapnia appears as the primary factor causing cerebrovascular vasoconstriction and decrease in cerebral blood flow with core heating.³⁸ Interestingly, and as addressed by others,^{22,37} there appears to be a core temperature threshold occurring somewhere between +1°C and +1.5°C that triggers heat-induced hyperventilation, hypocapnia, and subsequent decrease in cerebral blood flow. Our data provide further evidence to the existence of this threshold given that T_{gi} rose by ~+1.3°C during HWI₄₁ but only by ~+0.6°C during HWI₄₀. Recent data suggest that such an acute cerebrovascular challenge appears safe for the brain parenchyma, at least in young male cohorts.²¹ However, it is known that central perfusion pressure (i.e., MAP) may be challenged by a decrease in SVR (coupled with sweating) to support heat dissipation. Consequently, when heat stress overlays with an additional cardiovascular

challenge, such as standing to exit of the bath, cerebral perfusion may be compromised, which would ultimately increase orthostatic intolerance.²³ Therefore, the relevant thermal modality to enhance muscle recovery, that is, HWI₄₁, might be at a greater risk than HWI₄₀ in developing pre-syncopal symptoms.

4.1 | Limitations

A number of limitations have to be considered in this study. In our view, the main one is the lack of continuous intramuscular temperature recording. A simultaneous recording of both core and intramuscular temperature would have offered the opportunity to partly elucidate whether one or both temperatures were determinant for muscle recovery. Finally, it is worth noting that the present study was executed in men. It is suggested that women receiving the same exogenous heat stress as men experience greater rise in core temperature due to their differences in physical characteristics (i.e., smaller body mass and higher fat mass) and sweat rate.³⁹ In addition, when environmental temperature is higher than skin temperature (as in the present study), women will gain heat faster than men due to their higher surface area-to-mass ratio.⁴⁰ Taken together, kinetics of core body temperature in response to the two HWI modalities from the present study should not be extrapolated to a cohort of women.

5 | CONCLUSION

The major finding of this study was that a single session of HWI₄₁ but not HWI₄₀ provided the required endogenous heat strain to mitigate muscle fatigue and soreness following mild EIMD. The data collected on this population of physically active men suggest that maintaining a core body temperature within the range of 38.5°C–39°C may be advantageous for muscle recovery. Further research is required to elucidate the potential relationships between HSP, core and muscle temperature, and the restoration of muscle function.

5.1 | Perspectives

From a field perspective, the current study highlights that lower-body HWI appears as a valuable intervention for athletes who require rapid muscle recovery between bouts of strenuous exercise. However, there may be logistical limitations in maintaining water temperature during immersion. Once the subjects were immersed,

heat exchange between the water and the body caused the water temperature to decrease below the desired target. This phenomenon occurred despite the use of a dedicated system designed to maintain a constant water temperature. Consequently, frequent monitoring of the water temperature became necessary, and additional hot water from an external source (e.g., tap) was required to sustain the target temperature. Taken together, it is evident that HWI may not be as practical as initially thought when aiming to maintain a specific target temperature within a narrow range, such as 41°C.

Achieving the core body temperature necessary for optimal muscle recovery may induce a transient disturbance in pre-frontal cortex oxygenation toward the end of immersion, warranting further investigations. Finally, it remains to be determined whether this “optimal” endogenous heat dose applies to women and purposes other than muscle recovery.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Benoît Sautillet  <https://orcid.org/0000-0003-4036-9622>

Grégoire P. Millet  <https://orcid.org/0000-0001-8081-4423>

Maryne Cozette  <https://orcid.org/0000-0002-8211-7151>

Guillaume Costalat  <https://orcid.org/0000-0003-0478-2882>

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